

ENTERPRISE REACT HEALTH CHECK

"Is Your Frontend Bleeding Money? A 5-Minute Audit for CTOs."

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1. Architecture & Scalability

Goal: Ensure new features can be added without breaking the legacy system.

 CRITICAL QUESTION (YES/NO)	 BUSINESS IMPACT
1. When adding a new feature, does your team modify code in 3+ different places?	Increased Time-to-Market: High Coupling slows down feature releases. Competitors will outpace you while your team is stuck untangling code.
2. Is core Business Logic tightly coupled inside UI Components?	Vendor Lock-in & Refactoring Costs: You cannot reuse logic for Mobile Apps or new interfaces. Rebuilding will be expensive and risky.
3. Is your folder structure organized by "Domain/Feature" rather than file type?	Onboarding Costs: New developers take 2 months to understand the codebase instead of 2 weeks. Cognitive load is too high.
4. Do you have a Code Splitting strategy per Route/Page?	Bounce Rate: Users land on the "Home" page but are forced to download the "Admin" code. A 1s delay can cost 7% in conversions.
5. Does the team have a standard workflow (like Nx Monorepo) for code sharing?	Systemic Risk: Without standardization, teams copy-paste code. A single bug requires fixing in 10 different places.

2. Performance & User Experience

Goal: Determine if the application is driving customers away due to latency.

 CRITICAL QUESTION (YES/NO)	 BUSINESS IMPACT
1. Are your Core Web Vitals (LCP, CLS) green (>90) on Mobile?	Higher Ad Costs (CPC): Google penalizes slow websites with lower SEO rankings and higher advertising costs.
2. Are you bundling heavy libraries (Lodash, Moment.js) into the Main Bundle?	Losing 3G/4G Users: Bloated bundles make the app unusable on slower networks, alienating a large segment of users.
3. Does the UI freeze (Lag) when handling large lists (> 1000 items)?	Operational Waste: For internal tools, a 5-minute lag per day x 100 employees equals thousands of wasted billable hours annually.
4. Are images automatically optimized (WebP/AVIF) based on the device?	Inflated Cloud Bills: Serving 4K images to mobile devices wastes bandwidth and increases AWS/Azure costs unnecessarily.
5. Do you use Client-side Caching (React Query/SWR) to avoid redundant API calls?	Server Overload: Redundant requests increase infrastructure costs and risk server downtime during peak traffic.

3. Security & Risk Management

Goal: Protect the business from legal liabilities and data breaches.

 CRITICAL QUESTION (YES/NO)	 BUSINESS IMPACT
1. Are Access Tokens stored directly in <code>localStorage</code> ?	Security Vulnerability (XSS): Hackers can inject malicious scripts to steal tokens and hijack Admin accounts.
2. Are dependencies scanned for vulnerabilities (npm audit) regularly?	Supply Chain Attack: Using outdated libraries is like leaving the back door unlocked. Customer data is at risk.
3. Are sensitive API Keys inadvertently exposed in the frontend code?	Financial Loss: Exposed keys (Google Maps, AWS) allow hackers to use your quota, causing massive billing spikes.
4. Does the app auto-logout when tokens expire or upon inactivity?	Compliance Failures: Failure to meet security standards (ISO, GDPR) can result in the loss of major B2B contracts.

 CRITICAL QUESTION (YES/NO)	 BUSINESS IMPACT
5. Is User Input sanitized before rendering?	Brand Reputation: Unsanitized inputs allow malicious code injection, potentially defacing your website or scamming users.

4. Code Quality & Team Health

Goal: Evaluate the sustainability and maintainability of the engineering team.

 CRITICAL QUESTION (YES/NO)	 BUSINESS IMPACT
1. Does the project enforce TypeScript with strict: true ?	Bug Fix Costs: Without strict types, bugs are found by customers in Production rather than by developers during coding.
2. Is there a CI/CD pipeline running automated tests before Merging?	Deployment Fear: Teams hesitate to release on Fridays due to fear of breaking the build, reducing business agility.
3. Is a strict Linter/Prettier configuration enforced codebase-wide?	Internal Conflict: Senior Developers waste time debating syntax formatting instead of solving business problems.
4. Is there live documentation (Storybook) for UI Components?	Resource Waste: Designers redraw buttons and Developers re-code existing components because they don't know they already exist.
5. Do you receive crash reports (Sentry) before customers complain?	Customer Churn: Users encounter errors and leave silently. If you don't track errors, you lose customers permanently.

AUDIT RESULTS

If you answered **"NO"** to any of the questions above, your Frontend system contains hidden "technical debt" that threatens revenue and scalability.

 **Don't let technical debt slow down your growth.** Contact me for a 30-minute "Refactoring & Optimization Roadmap" consultation.

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